

Lily Oforiwa Sam

lilyoforiwaas@gmail.com • (646) 708-0047 • [LinkedIn](#) • [GitHub](#)

EDUCATION

Columbia University in the City of New York
Computer Science

New York, NY
August 2019 - May 2023

Relevant Coursework: Intro to Java, Intro to Python for Engineers, Data Structures and Algorithms, Discrete Mathematics, Advanced Programming, Computational Linear Algebra, Databases, User Interface Design, Advanced Software Engineering

HONORS & AWARDS

- **John W. Kluge Scholarship** – awarded to top 5% of Columbia College Admitted students class.
- **Columbia College Dean's List** - awarded per semester for 3.6 or better GPA in 12 or more points of letter credit.
- **Code2040 Finalist (2021/2022)** - Chosen from 1,000+ applicants for highly competitive technical accelerator for Black and Latinx technologists.
- **Twitter DevelopHER 2022 Finalist** - 1 of 30 students selected from 1,200+ applicants; won the #ReadySetCode team programming challenge.

WORK EXPERIENCE

Deutsche Bank Global Technology
Software Engineer

Cary, NC
July 2023 – Present

Career Highlights

Instant Payments Tribe Promotion

March 2026

- **Promoted** from Analyst (Software Engineer I) to Associate (**Software Engineer II**) in recognition of consistent delivery of robust payment services and demonstrated success in mentoring junior engineers.

Instant Payments AI Innovation Day Hackathon

March 2026

- Developed AI Agent to automate OSS remediations by writing GitHub Actions workflow to use GitHub Copilot AI tool to access JFrog Xray scans of packaged code and remediate vulnerabilities.

Instant Payments Tribe Contributions

- Built multiple Java SpringBoot microservices to support new instant payments scheme for US and Brazil.
- Saved 20,000+ engineering hours annually by building robust CI/CD pipelines with automated workflows to check controls that enabled on-demand same-day production releases and eliminated manual release overhead.
- Integrated real time payment services with MongoDB Enterprise and Oracle ExaCC Database.
- Set up monitoring and observability for services and code components on Google Cloud Platform and New Relic.
- Led on building CI/CD pipelines to support GitHub-based deployment workflows across development, UAT, SIT, and PROD environments.
- Wrote Terraform scripts to provision Kubernetes resources on Google Kubernetes Engine (GKE) and GCP.
- Wrote OracleSQL stored procedures and functions to build payment processing channels for storing and auditing transactions.
- Built ETL pipelines to process static account and customer data for instant payments, and designed data replication architecture to isolate time-critical payment flow operations from non-time-critical operations.

Columbia University in the City of New York

Teaching Assistant – COMS 4111

New York, NY
Sep 2022 – May 2023

- Designed course in Python programming language to cover relational and non-relational databases and instructed over 450 students in a graduate-level databases course along with grading exams and assignments.
- Held weekly office hours in Python, Flask and taught databases in MySQL, MongoDB and Neo4j to assist students with understanding course concepts.

Columbia University Center for Student Advising
Tutor

New York, NY
Sep 2021 – May 2023

- Held online private tutoring sessions for students to help them better understand data structures in Python and Java.
- Assisted students with mastering their course content and maximizing their potential for academic achievement.

Amazon Inc.

Seattle, WA

Software Development Engineer Intern (Full Stack)

June 2022 – Aug 2022

- Developed payments service API enabling business enterprises and customers to access health insurance contract data.
- Built frontend portal with React and TypeScript using Amazon's Harmony framework and AWS Polaris library.
- Deployed and maintained payments web service on AWS infrastructure including ECS, Fargate, S3, and DynamoDB.

Sponsors of Educational Opportunity (SEO)

Remote

Tech Developer Intern

June 2021 – Aug 2021

- Built a full stack web app to query data from the New York Times API using Python and Flask.
- Implemented development, testing, and automation tools to support product infrastructure.

PROJECTS & CONTRIBUTIONS

Text-to-SQL LLM Calibration Evaluator

Python, SQLite, Pandas, Matplotlib, Groq API

- Engineered a multi-model evaluation pipeline to measure Expected Calibration Error (ECE) and confidence reliability across 5 open-source LLM families on the academic Spider Text-to-SQL benchmark, using the Groq API.
- Designed an execution-match testing pipeline using local SQLite databases to grade zero-shot AI-generated queries by result correctness, bypassing the limitations of standard string-comparison grading.
- Uncovered a systemic overconfidence pattern across all evaluated models (92–99% stated confidence vs. 75–88% actual execution accuracy), and identified family-dependent scaling effects on calibration quality.

Agenta (Open-Source LLM Engineering Platform)

Python, PostgreSQL, GitHub

- Merged PR #6092 to resolve issue #5934, extracting duplicated delivery-upsert logic from three DAO methods into a shared helper module to eliminate drift risk between conflict-target rules.
- Verified no behavioral regressions via full local test suite (10 tests across trigger-claim and delivery-lifecycle coverage) before and after refactor.
- PR was reviewed and merged by project maintainer with no correctness issues flagged.

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, JavaScript, TypeScript, HTML, CSS, React, AngularJS, Flask, Spring Boot, Maven, Apache Kafka, Bash

Cloud & DevOps: Google Cloud Platform (GKE, Cloud Storage, Secret Manager), AWS (ECS, Fargate, S3, DynamoDB), Terraform, Docker, Kubernetes, GitHub, New Relic, GitHub Actions

Databases: MySQL, MongoDB, Neo4j, Oracle SQL

Tools & Methodologies: JIRA, Git, Agile/Scrum, BDD (Gherkin, Cucumber)